

BLACKBLOX SDK - Quick Start Guide v0.2

Estimated setup time: 15-20 minutes.

1. Requirements

Hardware: BLACKBLOX Processor module, BLACKBLOX RGB Matrix module and a USB-C cable.

Software: Arduino IDE 2.x.

2. Install Arduino IDE

Download Arduino IDE from <https://www.arduino.cc/en/software> and install it with the default settings.

3. Install ESP32 Support

Open Tools -> Board -> Boards Manager, search for 'esp32' and install the package from Espressif Systems.

4. Install the BLACKBLOX SDK

On the supplied USB flash drive you will find the folder 'BLACKBLOX'. Copy this folder into:

Documents/Arduino/libraries/

After copying, restart Arduino IDE.

5. Connect the Hardware

Connect the BLACKBLOX Processor module to the RGB Matrix module and connect the processor to your computer using USB-C. The default matrix address is 0x10.

6. Select the Board

Select 'ESP32S3 Dev Module' and choose the correct COM port.

7. Run the First Example

Open File -> Examples -> BLACKBLOX -> Matrix -> BasicColors and upload it. The RGB Matrix should turn blue.

8. Try the Smiley Example

Open the Smiley example and upload it. A yellow smiley should appear.

9. Two Matrices

Configure two matrices with addresses 0x11 and 0x12 and run the DualMatrix example.

10. I2C Addresses

The BLACKBLOX SDK uses 7-bit I2C addresses. RealTerm displays the 8-bit transfer address. Example: SDK address 0x10 corresponds to 0x20 in RealTerm.